Appendix I — H2a Echo-Protected Hardware Test: The Fixed-Point Law Is Falsified on Hardware

UIP Phase 1 · Derek Hone, Remnant Fieldworks · July 12, 2026 · Preregistration: `prereg_h2a_echo_fixed_point.md` (committed before execution) · IBM Quantum job `d99h2nd2su3c739keu80` · Follows Appendix H (H1: FAIL, 383.7%)

Headline result: FAIL. Measured steady-state drift D* = 0.0550 versus predicted 0.0302 — relative deviation **81.9%**, far outside the preregistered ±10% window [0.0272, 0.0333]. Per the preregistration's interpretation rule, this failure falsifies **the law itself, not merely its inputs**. The idealized recirculation fixed-point law of Appendix A is a simulation result, not a hardware law. Published unedited at its preregistered threshold.

1. Preregistered hypothesis (H2a)

Appendix H (H1) rejected the fixed-point prediction at 383.7% deviation using bare idle delays; the recorded diagnosis was an input error — bare idles decay at T2* while device calibration publishes echo-T2. H2a tested the sharpened claim: with an X echo pulse at the midpoint of every idle delay (so decay is governed by the calibrated echo-T2), the identical law

$$C^* = f / (1 - (1-f)\cdot e^{-\gamma}), \quad \gamma = \text{delay} / \text{T2\_echo}, \quad D^* = (1 - C^*)/2$$

predicts hardware steady-state drift within ±10% (relative). Kill-condition and interpretation rules were committed before execution: *"FAIL: the law itself (not merely its inputs) is falsified on hardware. No post-hoc parameter adjustment counts as a result."*

2. Protocol

Item	Value
Backend / qubit	ibm_kingston (Heron r2, 156 qubits), qubit 140 — same device and qubit as H1
Calibration T2 (echo)	601.1 μs (day-of-run)
Cycle	40 μs idle per cycle, echo X at midpoint (delay/2 – X – delay/2), net flip undone; gain f = 0.5; 8 cycles
Recirculation model	Mixture-of-ages circuit family, weights $f(1-f)^{n-k}$; 4096 shots each; X-basis readout; $C = \|2p_0 - 1\|$
Part A prediction (on record)	γ = 0.0665, per-cycle decay = 0.9356, C* = 0.9395, **D* = 0.0302**, PASS window [0.0272, 0.0333]
Job ID	`d99h2nd2su3c739keu80`

3. Results (raw, unedited)

Cycle	C (measured)	D (measured)
0	0.9951	0.0024
1	0.9341	0.0330
2	0.9054	0.0473
3	0.9000	0.0500
4	0.8913	0.0544
5	0.8909	0.0545
6	0.8897	0.0551
7	0.8915	0.0543
8	0.8888	0.0556

Steady state (mean of last 3 cycles): D* = 0.0550. Predicted: 0.0302. Relative deviation: 81.9% → FAIL at the preregistered ±10%.

4. Comparison across both hardware runs

	H1 (bare idle)	H2a (echo-protected)	Predicted
Steady-state D*	0.1463	0.0550	0.0302
Relative deviation	383.7%	81.9%	—
Verdict	**FAIL**	**FAIL**	—
Stable flat steady state observed	**Yes**	**Yes**	Yes (structure)

5. What the failure teaches (recorded diagnosis, zero evidential weight)

- **The H1 diagnosis was substantially correct.** Echo protection reduced the deviation by roughly a factor of five (383.7% → 81.9%) and raised steady-state coherence from ≈0.71 to ≈0.89. The T2 vs T2* distinction is real and dominant in bare-idle circuits.
- **But correcting the dominant error was not sufficient.** The residual gap is attributable to loss channels absent from the two-parameter model: single-qubit gate error accumulated over the added H and X pulses (deep family circuits contain many additional gates), readout error, imperfect refocusing by a single midpoint echo relative to the multi-pulse sequences used in calibration, and amplitude damping (T1), which echoes do not remove.
- **The qualitative fixed-point structure survived both runs.** In H1 and H2a alike, coherence converged rapidly to a flat, stable steady state exactly as the recirculation model predicts — but at a level set by the device's true aggregate loss, not by calibration T2 alone.

> **Final Phase 1 status of the fixed-point law.** Confirmed quantitatively in simulation (<2%, Appendix A). Falsified quantitatively on hardware at ±10%, twice, at two levels of input honesty (Appendices H, I). Surviving content: the qualitative prediction of fast convergence to a stable coherence plateau under recirculation. A quantitative hardware law would require an extended model

> with explicit gate-error, readout-error, and T1 terms — that is a Phase 2 hypothesis, deliberately **not** attempted here. Per the program's no-third-attempt rule, Phase 1 hardware testing closes with this result.

6. Ledger after Appendix I

Appendix	Claim	Outcome
A	Recirculation fixed point (simulation)	**PASS (<2%)**
B (v1, v2)	Inheritance-guided budget allocation	**FAIL (×2)**
C	Restricted-access advantage	**PASS**
D–E	Crossover law s*(γ)	**PASS**
F	Near-optimality gauntlet	**FAIL (12.19%)**
G	Universality ("Unified")	**FAIL — retired**
H	Fixed-point law, calibration inputs, hardware	**FAIL (383.7%)**
I	Fixed-point law, echo-protected, hardware	**FAIL (81.9%)**

Phase 1 closes at 4 confirmed / 5 falsified — both hardware verdicts delivered by a device outside the author's control, both published unedited the same day.

7. Reproducibility

Preregistration commit, full script (`h2a_echo_test.py`), raw cycle data, and job ID `d99h2nd2su3c739keu80` are public in the repository (`github.com/derekhone/uip-phase1-testbeds`) and archived under DOI `10.5281/zenodo.21246246`. Anyone with an IBM Quantum open-plan account can rerun the test. Refutations and tightenings are solicited as explicitly as confirmations, via repository issues.

"A false balance is an abomination to the LORD, but a just weight is His delight." — Proverbs 11:1